

PRODUCT REQUIREMENTS DOCUMENT

AI & Humanity — Personal Course Portfolio Website

Version 1.0 | February 18, 2026 | Ready for Development

1. Vision & Purpose

Build a visually stunning, secure, single-page portfolio-style website for a student's AI & Humanity course. The site should feel like a premium personal site (think: a developer portfolio crossed with an interactive art piece) — but the content centers on AI & Humanity coursework, reflections, projects, and resources. The 3D visuals should reinforce the theme: the intersection of technology and the human experience.

The site should be architected so new features and pages can be added incrementally over the semester.

2. Tech Stack

Layer	Technology	Rationale
Framework	Next.js 14+ (App Router)	SSR/SSG, file-based routing, easy to extend
Language	TypeScript	Type safety, better developer experience
Styling	Tailwind CSS 3	Rapid, consistent styling
3D Graphics	Three.js + R3F + Drei	Declarative 3D in React, huge ecosystem
Animation	Framer Motion	Page transitions, scroll animations
Auth	NextAuth.js (Auth.js v5)	Pluggable providers, session management
Security	Helmet, CSP, Rate Limiting	Production-grade security headers
Deployment	Vercel	Zero-config for Next.js
Package Mgr	pnpm	Fast, disk-efficient

3. Site Architecture & Pages

3.1 Landing / Hero Section

- Full-viewport 3D scene as the background (see §5 for 3D spec)
- Floating title text: student name + “AI & Humanity” with subtle glow/distortion shader
- Smooth scroll-down indicator (animated chevron)
- Navigation bar: glassmorphism style, semi-transparent, sticky on scroll

3.2 About Section

- Split layout: stylized portrait or avatar on left, bio paragraph on right

- Scroll-triggered entrance animations (fade + slide via Framer Motion)
- Course Info card: course name, semester, institution, instructor

3.3 Projects Section

- Responsive card grid (1 col mobile → 2 col tablet → 3 col desktop)
- Each card: thumbnail, title, description, tags, 3D tilt hover effect
- Click expands to detail view or navigates to /projects/[slug] page
- Data source: local MDX or JSON files in /content/projects/

3.4 Reflections / Journal Section

- Blog-style list rendered from MDX files in /content/reflections/
- Each entry: date, title, estimated read time, preview snippet
- Full reading view with prose styling via @tailwindcss/typography

3.5 Resources Section

- Curated links to papers, books, and media grouped by category
- Categories: Ethics Frameworks, Landmark Papers, Documentaries, Tools

3.6 Contact / Footer

- Contact form (name, email, message) wired to serverless API route (/api/contact)
- Server-side validation + rate limiting
- Social links (GitHub, LinkedIn, email) and footer credits

4. Design System

4.1 Color Palette

Dark-mode-first design to let the 3D graphics pop.

Token	Value	Usage
--bg-primary	#0a0a0f	Page background
--bg-surface	#12121a	Cards, panels
--accent-primary	#6366f1 (Indigo)	Links, CTAs, glow effects
--accent-secondary	#8b5cf6 (Violet)	Gradient partner, hover states
--accent-cyan	#22d3ee	3D particle highlights, badges
--text-primary	#f1f5f9 (Slate 100)	Headings, body text
--text-muted	#94a3b8 (Slate 400)	Captions, metadata

4.2 Typography

Element	Font	Weight	Size
Headings	Inter / Space Grotesk	700	3xl–6xl
Body	Inter	400	base–lg
Code/mono	JetBrains Mono	400	sm

4.3 Effects & Micro-interactions

- **Glassmorphism** on nav and cards: backdrop-blur-xl, subtle border, low-opacity bg
- **Gradient text** on hero title: indigo → violet → cyan with bg-clip-text
- **Hover states:** all interactive elements have smooth 200ms transitions
- **Scroll animations:** sections fade/slide in using Framer Motion whileInView
- **Cursor glow** (optional): subtle radial gradient follows cursor on hero section

5. 3D Graphics Specification

This is the centerpiece. The 3D scene should feel premium and thematic — evoking the duality of AI and humanity.

5.1 Hero Scene — “Neural Constellation”

A full-viewport interactive 3D scene rendered with React Three Fiber:

Particle Field

- 2,000–5,000 particles representing a neural network / constellation
- Small glowing spheres using PointsMaterial or instanced meshes with custom shaders
- Color gradient from indigo → violet → cyan based on Y-position
- Subtle constant drift animation (sine-wave oscillation on position)

Connection Lines

- Dynamic lines drawn between nearby particles (distance threshold)
- Creates a network/graph effect using THREE.LineSegments with glow shader

Central Form

- Morphing icosahedron — vertices displaced by simplex noise over time
- Organic “breathing” feel with wireframe + solid hybrid and emissive edges
- Alternatives: DNA double helix of spheres, or human silhouette made of particles

Interactivity

- Mouse/touch parallax: camera subtly shifts based on pointer position
- On hover over central form: particles accelerate, colors shift
- On scroll: depth-of-field increases, particles spread into content below

Post-Processing

- Bloom: subtle, threshold ~0.8, intensity ~0.4
- Chromatic aberration: very subtle, offset ~0.002
- Vignette: mild darkening at edges

Performance

- Use InstancedMesh for particles (not individual meshes)
- Responsive particle count: 5,000 desktop → 2,000 mobile
- Target 60fps using useFrame delta for frame-rate-independent animation
- Canvas with dpr=[1, 2] for responsive pixel ratio
- Lazy load the 3D scene — show gradient placeholder until Three.js initializes

5.2 Section Dividers (Stretch)

- Thin horizontal 3D wave/terrain strip between sections using displaced plane geometry

5.3 404 Page (Stretch)

- Glitching 3D “404” text with particles flying off it

6. Content Management

All content lives in the repo as flat files — no CMS or database needed for v1.

To add a new project or reflection, just drop a new .mdx file in the right folder and it appears on the site.

File Structure

```
/content
├── projects/
│   ├── project-1.mdx
│   └── project-2.mdx
├── reflections/
│   ├── week-1-what-is-ai.mdx
│   └── week-2-ethics-of-automation.mdx
└── resources.json
```

MDX Frontmatter (Projects)

title, date, tags[], thumbnail (optional), summary

MDX Frontmatter (Reflections)

title, date, tags[], readTime

7. Security Requirements

Build with real security practices — it demonstrates competence and is good habit.

7.1 HTTP Security Headers

Set via next.config.js headers():

Header	Value
Content-Security-Policy	default-src 'self'; script-src 'self' 'unsafe-eval'; style-src 'self' 'unsafe-inline'; img-src 'self' data: blob:; frame-ancestors 'none'
X-Content-Type-Options	nosniff
X-Frame-Options	DENY
Referrer-Policy	strict-origin-when-cross-origin
Permissions-Policy	camera=(), microphone=(), geolocation=()
Strict-Transport-Security	max-age=63072000; includeSubDomains; preload

7.2 API Route Protection

- **Rate limiting** on /api/contact: max 5 requests per IP per minute
- **Input validation:** sanitize all form inputs server-side using Zod schema validation
- **CSRF protection:** SameSite=Strict cookies if adding auth later

- **Environment variables:** all secrets in `.env.local`, never committed

7.3 Dependency Security

- `pnpm audit` in CI pipeline
- Pin major versions in `package.json`
- `.gitignore` covers `.env*`, `node_modules`, `.next`

7.4 Auth (Future-Ready)

- NextAuth.js configured with placeholder provider (GitHub OAuth or email magic link)
- Auth is off by default for v1 (public site), can be flipped on to gate routes
- Middleware in `middleware.ts` to protect `/admin/*` routes when enabled

8. Performance Requirements

Metric	Target
Lighthouse Performance	≥ 90
First Contentful Paint	< 1.5s
Largest Contentful Paint	< 2.5s
Cumulative Layout Shift	< 0.1
3D Scene FPS	≥ 60fps desktop, ≥ 30fps mobile
Bundle Size (JS)	< 300KB gzipped (excl. Three.js lazy chunk)

Performance Strategies

- Lazy-load the Three.js canvas with next/dynamic and ssr: false
- Use Suspense with a gradient fallback while 3D loads
- Static-generate all content pages at build time (generateStaticParams)
- Optimize images with next/image
- Tree-shake Three.js imports

9. Accessibility

- All text content meets WCAG 2.1 AA contrast ratios
- Semantic HTML: <nav>, <main>, <article>, <section>, <footer>
- Skip-to-content link
- Keyboard navigable: all interactive elements focusable with visible focus rings
- aria-label on icon-only buttons/links
- prefers-reduced-motion: disable 3D animations, use static gradient fallback
- Form inputs have associated <label> elements

10. Responsive Breakpoints

Breakpoint	Width	Adjustments
Mobile	< 640px	Single column, 2K particles, smaller headings, hamburger nav
Tablet	640–1024px	2-column project grid, medium particle count
Desktop	> 1024px	Full layout, 5K particles, all effects enabled

11. Development Phases

Phase 1 — Foundation (This Build)

- Next.js project scaffolding with TypeScript + Tailwind
- Design system: colors, typography, global styles
- Navbar + Footer
- Hero section with full 3D “Neural Constellation” scene
- About section
- Projects section (card grid + 2–3 placeholder projects)
- Reflections section (list + 1–2 placeholder entries)
- Resources section
- Contact form with API route + validation + rate limiting
- Security headers
- Responsive + accessible
- Deploy to Vercel

Phase 2 — Content & Polish (Future)

- Real course content in MDX
- Project detail pages with rich MDX rendering
- Reflection detail pages
- Search / tag filtering
- Dark/light mode toggle
- Reading progress bar on reflections
- RSS feed for reflections

Phase 3 — Interactive Features (Future)

- Authentication (NextAuth) for admin/draft mode
- Comment system on reflections (e.g., Giscus or custom)
- AI chatbot widget (Anthropic API) for visitors to ask about course content
- Interactive AI demos embedded in project pages
- Analytics dashboard (Plausible or Vercel Analytics)

12. Implementation Notes for Claude Code

1. Start with `pnpm create next-app` using App Router, TypeScript, Tailwind, and ESLint
2. Install 3D deps: `three`, `@react-three/fiber`, `@react-three/drei`, `@react-three/postprocessing`, `@types/three`
3. Install animation: `framer-motion`
4. Install utilities: `zod` (validation), `next-mdx-remote` or `@next/mdx` (MDX support)
5. 3D scene must be a client component ('use client') loaded via `next/dynamic` with `ssr: false`
6. Use `simplex-noise` for the morphing icosahedron vertex displacement

7. All color tokens as CSS custom properties in `globals.css`, mapped in `tailwind.config.ts`
 8. Test on mobile — 3D scene should gracefully degrade (fewer particles, no post-processing)
 9. Ship placeholder content — realistic dummy text so the site looks complete on day one
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This PRD is designed to be handed directly to Claude Code. Every section is actionable. Build Phase 1 in full, and structure the codebase so Phases 2 and 3 can be added without refactoring.